

Md. Safirur Rashid

☎ +880-1815-777772 | ✉ md.safirur.rashid@gmail.com | [in md-safirur-rashid](#) | [5AF1](#)

Middle Badda, Dhaka-1212

About Me

Passionate ML Engineer, skilled in Computer Vision & Data Science, eager to use deep learning to build impactful solutions at scale. As a lifelong learner eager to expand my knowledge in collaborative environments, I aim to drive innovation and tackle complex challenges in the field of Data Science and Machine Learning.

Experience

AlterSense Ltd.

Machine Learning Engineer

Tejgaon, Dhaka

April. 23 – Present

- ▶ As a part of the ML team of the company working on Retail Analytics extraction from CCTV camera footage.
- ▶ Currently working on exploiting biases to reduce computational cost of the ML pipeline.
- ▶ Work closely with the operations team during installation and maintenance of the cameras.
- ▶ Implemented lens curvature elimination in preprocessing steps of the pipeline to align multiple camera and track customers position for analytics.

Pioneer Alpha

Machine Learning Engineer

Elephant Road, Dhaka

Aug. 22 – Mar. 23

- ▶ As part of the Computer Vision team, improved an already deployed traffic detection model accuracy by extracting insights from research papers and retraining with new field data and misclassified data.
- ▶ Worked as an intern for the company for the first 4 months. [\[Verify\]](#)
- ▶ Developed video curriculum for Amar iSchool's "Research Paper Code Implementation" course, empowering learners to re-implement state-of-the-art AI research and deploy practical models.
- ▶ Enhanced learning through interactive demonstrations showcasing model deployment and real-world impact.

Omdena

Machine Learning Engineer [\[Verify\]](#)

BigCodeGen: Predicting Traffic flow congestion on Urban Motorway Networks [\[Verify\]](#)

Dec. 22 – Mar. 23

- ▶ Lead the ML Model team of 7 international collaborators for improving the core image classification model used to classify the congestion category and creating automated data labeling pipeline.
- ▶ Utilized YOLOv7 & v8 for vehicle detection, & lane segmentation, to automate training data generation and integrated traffic light detection to pinpoint congested intersections, ensuring accurate traffic flow analysis.
- ▶ In these 3 months we achieved a 6% field data accuracy boost through targeted training refinements guided by Exploratory Data Analysis as documented in the final [report blog](#), a collaborative effort I was proud to be a part of.

Bôndy: Using Satellite Imagery for Reforestation in Madagascar [\[Verify\]](#)

May. 22 – Aug. 22

- ▶ Worked with 20 international collaborators on a 3-month long project to enhanced geospatial imagery analysis via utilizing the SRCNN super-resolution model to improve satellite and drone image quality for tree counting.
- ▶ Increased tree counting model accuracy by 3% and implemented user experience improvements within the analysis pipeline as detailed in our final [report blog](#).

Education

Islamic University of Technology

Bachelor of Science in Computer Science and Engineering

Boardbazar, Gazipur

Jan. 2018 – May. 2022

- ▶ CGPA : 3.67 (Out of 4)

Bir Shreshtha Noor Mohammad Public College

Secondary and Higher Secondary

Peelkhana, Dhaka

Jan. 2004 – Dec 2016

- ▶ Higher Secondary Certificate, GPA - 4.92 (Out of 5)

Test Scores

International English Language Testing System (IELTS)

May, 2024

Overall Band Score: 7.5 CERF Level: C1 [\[Verify\]](#)

- Listening: 8.5 Reading: 8.0 Writing: 6.5 Speaking: 7.5

Graduate Record Examinations (GRE)

Mar. 2024

Combined Score: 307 [\[Verify\]](#)

- Verbal Reasoning: 148 Quantitative Reasoning: 159 Analytical Writing: 3.5

Certifications

Data Science Bootcamp | by Upskill [\[Verify\]](#)

Dec 15, 2021

Deep Learning Specialization | by Deeplearning.ai [\[Verify\]](#)

Nov 22, 2020

Simulation and modeling of natural processes | by University of Geneva [\[Verify\]](#)

May 19, 2020

Machine Learning | by Stanford University [\[Verify\]](#)

Apr 22, 2020

Academic Research and Projects

Joint Pose Based Sign Language Translation Using Graph Convolutional Network

Dec 2020 – May 2022

- Teamed up with 3 researchers under Dr. Md. Hasanul Kabir to develop a novel approach for extracting spatio-temporal features using Human Pose Estimation.
- Leveraged pose estimation methods to generate robust features for Transformer models, enhancing the accuracy of existing Sign Language Translation models, particularly on the PHOENIX14T dataset.
- For a comprehensive understanding of the methodology and findings presented in our thesis, kindly refer to the final defense report available [here](#).

Pathshala | Flutter, Dart, Python, Fast.ai, Tflite | [View on !\[\]\(f507db636256ac11a5525ef93ec6b8d7_img.jpg\)](#)

Oct 2021

- Collaborated with a team to develop a Flutter mobile app for kindergarteners, teaching them to write Bangla and English letters through interactive drawing and AI-powered recognition.
- Spearheaded the deep learning aspect, creating six distinct models using MobileNetV3 architecture within the fast.ai library. These models analyze children's drawings and provide real-time feedback on letter accuracy
- Efficiently trained and exported the models to tflite files for seamless integration into the Flutter app, empowering kids to learn through playful exploration.

Publications

- **Rashid, M. S.**, Morshed, M. S., Islam, M. U., Rashid, S., Mahmud, A., & Islam, A. (2024, May). Mycological Examination of Microscopic Fungi Images with Deep Learning and Gradient Weighted Class Activation Mapping Visualization. Paper Accepted for Publication and Presented at the 2024 International Conference on Bioengineering Engineering - Advances in Science and Engineering Technology (ASET) International Conferences, Dubai, UAE. Manuscript in preparation.

Achievements

International Rover Design Challenge 2021

Sept 06, 2021

IUT Mars Rover- Team Avijatrik [\[Verify\]](#)

Boardbazar, Gazipur

- We secured 11th position among the total of 32 teams in terms of overall performance, innovative execution and total score points.
- We were the only team from Bangladesh to achieve such a front line position in the competition

Skills

Python | PyTorch | W&B | Fast.ai | Scikit-Learn | Threading | OpenCV | NumPy | SciPy | PyMC | ArviZ | Pandas | Plotly
C++ | Conan | TensorRT | ONNX Runtime | TFLite | Docker | Redis | Kafka | MySQL | Julia | Octave | Streamlit | FastAPI
VS Code | PyCharm | Git | GeoGebra | Loguru | Addict | Timm | EasyFSL | Ultralytics | $\LaTeX$ | Pathlib | Mamba | mpmath